

IPCC Tier 2 Livestock GHG Uncertainty Calculator

Pre-Flight Questionnaire — GMH Uncertainty Translator

CIAT / CGIAR Alliance — ClimateActionNetZero

Fill this out **before** opening the Translator on claude.ai. When you start the chat, paste the contents of this filled page as your first message — Claude will use it to skip the warm-up questions and go straight to inspecting your data.

Save this file as `my_inventory_questionnaire.docx` and type your answers directly into the form below. Italic lines are guidance and can be left in place.

1. Country and inventory year

- **Country:** _____
- **Continental region** (tick one — drives the IPCC BW deviation benchmark): ☐ africa ☐ asia ☐ europe ☐ americas ☐ oceania ☐ global
- **Inventory year being prepared:** _____

2. IPCC guidelines version

Tick one:

- ☐ **IPCC 2006 Guidelines** (still the default reporting requirement in many countries)
- ☐ **IPCC 2019 Refinement** (newer; updated emission factors, more manure systems)

If unsure, ask your national focal point or pick 2006 — the app accepts both.

3. Cattle sub-categories

List the sub-categories your inventory splits cattle into and the approximate head count of each. You can use any names — Claude will map them to the app's controlled vocabulary.

Sub-category name	Head count	Sex (M/F/mix)	Age class (adult / 1–3yr / calf)
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Example row: dairy cows / 500,000 / F / adult

4. Manure management

For each sub-category, roughly what % of manure goes to each system? Per sub-category, the percentages must sum to 100. Recognised systems:

- pasture, daily_spread, solid_storage, solid_storage_covered*, dry_lot, deep_bedding
- liquid_slurry, lagoon, composting, anaerobic_digester*, aerobic_treatment*, burned_for_fuel*

**available only when IPCC version = 2019 Refinement.*

Sub-category	System 1 (%)	System 2 (%)	System 3 (%)	System 4 (%)
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Example: dairy cows: pasture 40%, solid_storage 35%, liquid_slurry 15%, lagoon 10% (sums to 100).

Optional — is the allocation itself uncertain? If yes, give a range per share (e.g. “pasture 40 %, lower 30 %, upper 55 %”). The app samples each MMS within its range and renormalises every Monte Carlo iteration so the totals still sum to 100 %. Leave blank if you are confident in the central allocation.

Sub-category	System	Central %	Lower %	Upper %
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5. Data fields you have

Tick the parameters for which you have country-specific values. Unticked parameters will be filled with IPCC defaults and flagged amber in the app's QA/QC tab for review.

Activity data (the population number)

- ☐ Animal population (head count) — **strongly recommended**

Animal characteristics

- ☐ Body weight (kg) — **BW**
☐ Mature body weight (kg) — **MW**
☐ Daily weight gain (kg/day) — **WG**
☐ Fraction of females pregnant in the year (includes pregnant heifers) — **pct_pregnant**

Production (dairy only)

- ☐ Daily milk yield per lactating cow (kg/day) — **Milk**
☐ Milk fat content (%) — **Fat**
☐ Milk protein content (%) — **MilkPR**

Feed

- ☐ Digestible energy of feed (% of gross energy) — **DE**
☐ Crude protein in feed (%) — **CP**

Climate (optional)

- ☐ Mean winter temperature (°C) — **Tw**

IPCC equation coefficients (*rarely measured locally; leave unticked unless your country has a published value*)

- ☐ **C_fi, C_a, C, C_p, Y_m, B_o, A_{SH}, U_E**
☐ Pasture/indirect N₂O emission factors (EF3_PRP, EF4, EF5, Frac_GASM_PRP, Frac_LEACH_PRP)

*Managed-storage manure-N₂O values (the direct EF and the volatilisation / leaching fractions) are entered **per manure-management system** in the Manure Management section above, not as single Parameters values.*

6. Where do your uncertainty estimates come from?

Tick one (or combine if different sources for different parameters):

- ☐ **(a) No uncertainty estimates** — use IPCC suggested $\pm\%$ from the catalogue
☐ **(b) Expert judgement $\pm\%$** — I'll write a single $\pm\%$ next to each value
☐ **(c) Measured confidence intervals** — I have explicit lower / upper bounds (or $\pm$ with

sample size)

- ☐ (d) **A mix** — I'll specify per parameter

7. Time series (optional)

- ☐ I have annual data for **5 or more years** that the app can use to estimate correlations between parameters.

• **Earliest year:** _____ **Latest year:** _____

• **Parameters covered (tick all that apply):**

- ☐ N
- ☐ BW
- ☐ MW
- ☐ WG
- ☐ Milk
- ☐ Fat
- ☐ pct_pregnant
- ☐ DE
- ☐ CP
- ☐ MilkPR

8. File(s) you'll upload

List the file names and a one-line description of what's in each.

1. _____
2. _____
3. _____

9. Anything else you want Claude to know

Free-form note: anything unusual about your data, sub-category definitions specific to your country, or known quirks. Optional.

That's it. Paste this filled page into the Translator and attach your data files. Claude will take it from here.